

Mohamed Ahmed Mostafa

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EDUCATION

B.S in Computer Science, Zagazig University, Egypt Oct. 2022 – Jun 2026
• **GPA:** 3.2/4.0 **Relevant Coursework:** Data Structures & Algorithms, OOP, Databases, OS, Distributed Systems, Compiler Design, Software Architecture, Big Data.

LEADERSHIP

Technical Instructor & Mentor 2025 – Present
ICPC Zagazig Community Egypt
• Mentoring **100+** students on Advanced DSA, with focus on **Dynamic Programming** and **Graph Theory**.

OPEN SOURCE CONTRIBUTIONS

Oppia 🐙 — *Open-source educational platform (6.6k ★, Google-supported nonprofit)* Jan 2026 – Present
• Fixed a **double confirmation modal** in Serial Chapter Mode by refactoring the story-editor navbar to call the publication service directly, eliminating a redundant UI layer. **#24508 #24527**
• Resolved a **WCAG 3.1.1 accessibility violation** by adding the missing **lang="en"** attribute to base HTML templates, fixing screen reader and SEO correctness before JS hydration. **#24923**
• Resolved **WCAG AA contrast violations** and removed invalid **ARIA menu roles** in the navigation bar to correct cross-browser accessibility bugs. **#25432**
• Eliminated a **flaky CI test** in the subprocess suite by scoping assertions to **stdout**, removing intermittent failures from gRPC background thread noise under CI resource contention. **#25223 #25061**
Lichess (lila) 🐙 — *Open-source chess server (17.9k ★)* Dec 2025
• Removed a **legacy dead-code endpoint** (/api route) causing white screens in dev environments, superseded by a reverse-proxy-served API docs route. **#18812**

PROJECTS

Bosla — **AI Learning Platform** 🐙 | **API** 🐙 | **Pipeline** 🐙 | *.NET 10, ASP.NET Core, EF Core, ONNX Runtime, SQL Server* 2025 – P
• Architected a **REST API** using **Clean Architecture** (Onion) with the Specification Pattern, fully decoupling business logic across Domain, Service, and Persistence layers.
• Built a real-time **AI inference pipeline** over WebSockets with **ONNX Runtime**; integrated Google Gemini for personalized learning roadmap generation.
• Deployed with **Docker** on SQL Server via EF Core, ensuring reproducible and environment-parity builds.
DevDiag — **Linux Diagnostic Tool** 🐙 | *Go, eBPF, Linux CLI/TUI* 2026
• Developed a low-overhead Linux CLI/TUI diagnostic tool utilizing **eBPF** for runtime syscall tracing and interception.
• Implemented an automated system including a **static audit** engine and **AI-powered fixing** mechanics to analyze and fix runtime anomalies.
BlitzMate Chess Engine 🐙 | *Python, Bit Manipulation, Algorithms* 2025 – Present
• Built a high-performance engine using **Bitboards**, **Alpha-Beta Pruning**, **Negamax**, and **Zobrist Hashing** to maximize move-selection depth via efficient bit manipulation.
Sarab — **Tunneling Automation CLI** 🐙 | *.NET 10, Cloudflare API, SQLite* 2026
• Automated DNS and ingress orchestration for local-to-cloud exposure, cutting manual configuration overhead by **40%**.
• Shipped as a **zero-dependency, self-contained binary** with SQLite-backed multi-identity state and secure SSH tunneling for public-IP-free remote access.
Skyra — **Weather Prediction** 🐙 | *Python, FastAPI, NASA API* 2025
• Built a low-latency **FastAPI** backend integrated with **NASA Open APIs** for real-time environmental data processing and weather prediction.

TECHNICAL SKILLS

Languages: C#, C++, Python, TypeScript, Go, SQL, HTML/CSS
Technical: OOP, Data Structures & Algorithms, Clean Architecture, Design Patterns, RESTful APIs, CI/CD, Testing
Frameworks: ASP.NET Core, EF Core, Dapper, ONNX Runtime, FastAPI
Databases & Cloud: SQL Server, PostgreSQL, Redis, Azure Cosmos DB, Azure, SQLite
Developer Tools: Git, GitHub, Docker, Linux, eBPF

ACHIEVEMENTS & COMMUNITY

Competitive Programming: **350+** problems solved on Codeforces/LeetCode (**Handle: Medo00**); active ICPC competitor.
Technical Content: Founder of “**DarkCipher00**” (YouTube).
Chess: **Top 5%** globally (**1700+** rating) on Chess.com.